



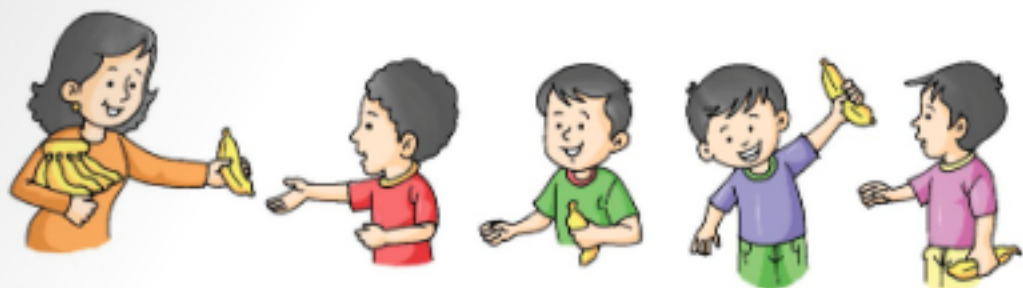
'Division' means 'equal sharing'.

Division as Distribution

Let us distribute 12 bananas equally among 4 boys.

Suppose we have a bunch of 12 bananas and there are 4 boys.

From this bunch, give 1 banana to each boy.



Now, the remaining number of bananas in the bunch is 8.

From this bunch, give 1 more banana to each boy.



Now, the remaining number of bananas in the bunch is 4.

From this bunch, give 1 more banana to each boy.



Now, no banana is left in the bunch.

Each boy gets 3 bananas.

Thus, we have divided 12 bananas into 4 equal parts and each part consists of 3 bananas.

We read it as : 12 divided by 4 is equal to 3.

We write it as : $12 \div 4 = 3$.

The symbol ' $\div$ ' stands for **division**.

In the division sum: $12 \div 4 = 3$, we call

12 as **dividend**, 4 as **divisor** and 3 as **quotient**.

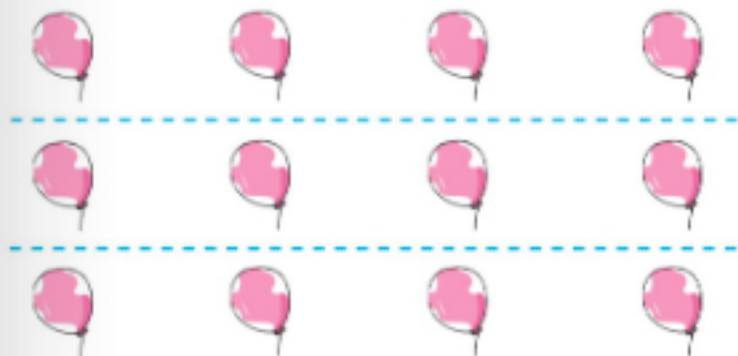
Thus, in a division sum. The

- number to be divided is called the **dividend**.
- number by which we divide is called the **divisor**.
- answer obtained on division is called the **quotient**.

Division as Forming Groups

We may take $12 \div 4$ as how many fours are there in 12.

Let us take 12 balloons and divide them into groups of 4 each.



Clearly, there are 3 groups.

How many fours are there in 12? Clearly, 3.

Hence, $12 \div 4 = 3$.

Take another example.

Suppose we have to find $15 \div 3$.

Here, we find how many threes are there in 15.

Let us take 15 leaves and divide them into groups of 3 each.



Clearly, there are 5 groups.

How many threes are there in 15? Clearly, 5.

Hence, $15 \div 3 = 5$.

Here, **dividend** = 15, **divisor** = 3 and **quotient** = 5.



Division as Repeated Subtraction

Suppose we have to divide 6 by 2.

Let us take 6 objects. Go on subtracting 2 as shown below.

Total number of objects = 6

Take away 2

Number of remaining objects = 4

Take away 2

Number of remaining objects = 2

Take away 2

Number of remaining objects = 0

How many times 2 has been subtracted? Clearly, 3 times.

How many twos are there in 6? Clearly, 3.

Hence, $6 \div 2 = 3$.

$$\begin{array}{r} 6 \\ - 2 \\ \hline 4 \\ - 2 \\ \hline 2 \\ - 2 \\ \hline 0 \end{array}$$



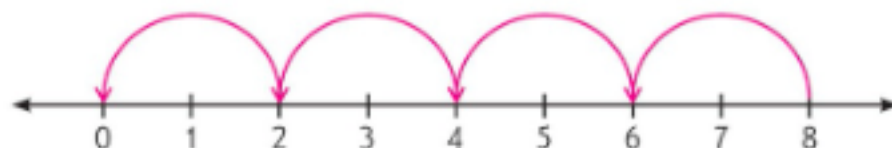
Division on the Number Line

Example 1: Use number line to find $8 \div 2$.

Solution: To find $8 \div 2$ means to find how many twos are there in 8.

Look at the number line given below.

Start from 8 and take back moves of 2 each to reach 0, as shown below.



How many moves of 2 each? Clearly, 4.

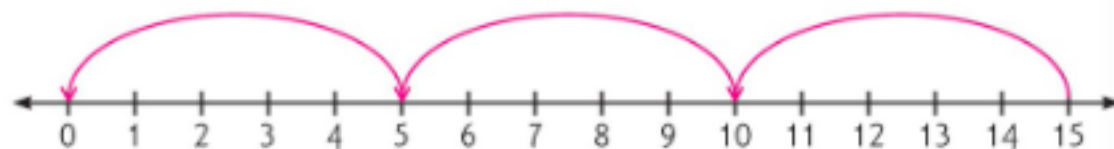
$$\therefore 8 \div 2 = 4$$

Example 2: Use number line to find $15 \div 5$.

Solution: To find $15 \div 5$ means to find how many fives are there in 15.

Look at the number line given below.

Start from 15 and take back moves of 5 each to reach 0, as shown below.



How many moves of 5 each? Clearly, 3.

$$\text{So, } 15 \div 5 = 3.$$



Exercise 25

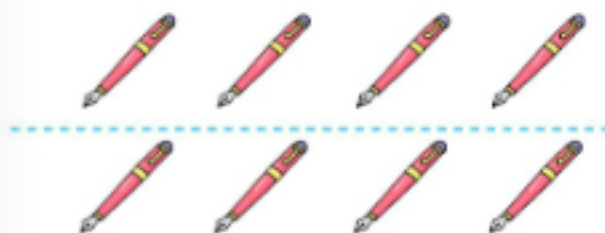
Look at the groups of objects and fill in the blanks.

1.



- (a) How many pineapples are there in all?
- (b) How many pineapples are there in each group?
- (c) How many groups are formed?
- (d) How many twos are there in 10?
- (e) $10 \div 2 =$

2.



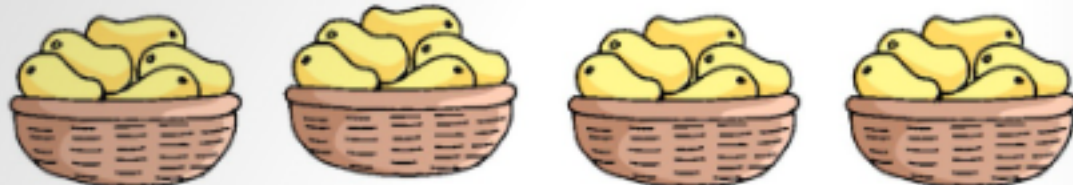
- (a) How many pens are there in all?
- (b) How many pens are there in each group?
- (c) How many groups are formed?
- (d) How many fours are there in 8?
- (e) $8 \div 4 =$

3.



- (a) How many flowers are there in all?
- (b) How many flowers are there in each pot?
- (c) How many pots are there?
- (d) How many threes are there in 18?
- (e) $18 \div 3 =$

4.



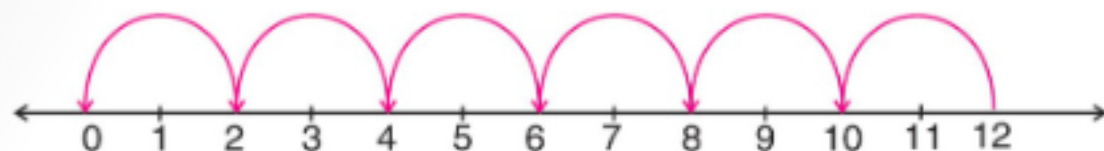
- (a) How many mangoes are there in all?
- (b) How many mangoes are there in each basket?
- (c) How many baskets are there?
- (d) How many fives are there in 20?
- (e) $20 \div 5 =$

Divide using the repeated subtraction method.

- | | | | |
|-----------------|-----------------|------------------|-----------------|
| 5. $9 \div 3$ | 6. $12 \div 3$ | 7. $16 \div 4$ | 8. $18 \div 6$ |
| 9. $21 \div 3$ | 10. $20 \div 5$ | 11. $24 \div 6$ | 12. $30 \div 5$ |
| 13. $24 \div 8$ | 14. $27 \div 9$ | 15. $40 \div 10$ | 16. $28 \div 7$ |

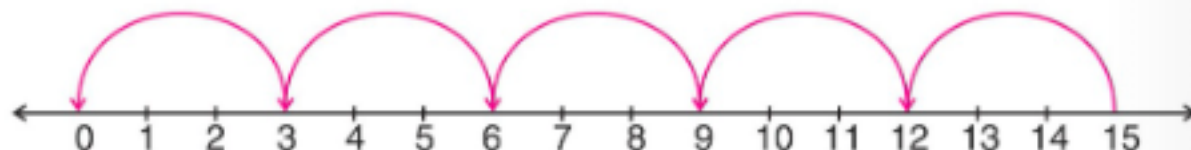
Look at the number lines given below and fill in the blanks.

17.



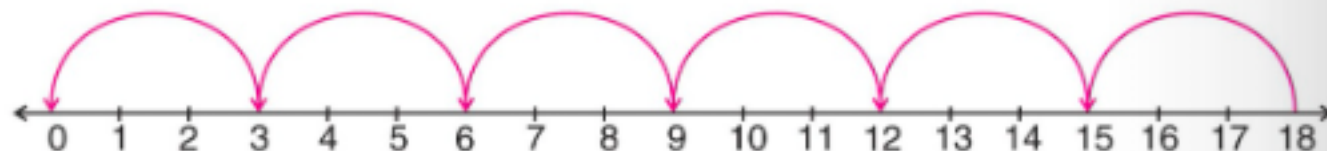
$12 \div 2 =$

18.



$15 \div 3 =$

19.



$18 \div 3 =$

20.



$16 \div 4 =$

Relation between Multiplication and Division

Let us take 4 bunches of 3 leaves each.

Total number of leaves = $4 \times 3 = 12$.

How many threes are there in 12? Clearly, 4.

$$\therefore 12 \div 3 = 4.$$

Again, divide the 12 leaves into 4 bunches.

How many leaves are there in each bunch? Clearly, 3.

$$\therefore 12 \div 4 = 3.$$

Thus, the multiplication fact $4 \times 3 = 12$ gives rise to two division facts:

$$12 \div 3 = 4 \text{ and } 12 \div 4 = 3$$

It is also clear that division is the inverse of multiplication.



Exercise 26

Write two division facts for each of the following multiplication facts.

- | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|
| 1. $3 \times 6 = 18$ | 2. $4 \times 5 = 20$ | 3. $7 \times 4 = 28$ | 4. $8 \times 3 = 24$ |
| 5. $9 \times 4 = 36$ | 6. $8 \times 5 = 40$ | 7. $4 \times 8 = 32$ | 8. $5 \times 6 = 30$ |
| 9. $7 \times 5 = 35$ | 10. $6 \times 7 = 42$ | 11. $7 \times 8 = 56$ | 12. $9 \times 7 = 63$ |
| 13. $10 \times 5 = 50$ | 14. $9 \times 1 = 9$ | 15. $1 \times 8 = 8$ | |

Write the multiplication facts of the following division facts.

- | | | | |
|---------------------|---------------------|---------------------|---------------------|
| 16. $24 \div 6 = 4$ | 17. $25 \div 5 = 5$ | 18. $21 \div 3 = 7$ | 19. $27 \div 3 = 9$ |
| 20. $48 \div 8 = 6$ | 21. $45 \div 9 = 5$ | 22. $54 \div 6 = 9$ | 23. $49 \div 7 = 7$ |
| 24. $9 \div 9 = 1$ | | | |

Properties of Division

We know that: (any number) $\times 1$ = number itself.

Let us study the division facts based on the multiplication fact as given below:

Multiplication Fact	Division Facts
$2 \times 1 = 2$	$2 \div 1 = 2$ and $2 \div 2 = 1$
$3 \times 1 = 3$	$3 \div 1 = 3$ and $3 \div 3 = 1$
$5 \times 1 = 5$	$5 \div 1 = 5$ and $5 \div 5 = 1$
$9 \times 1 = 9$	$9 \div 1 = 9$ and $9 \div 9 = 1$ and so on.



Thus, $2 \div 1 = 2$, $3 \div 1 = 3$, $5 \div 1 = 5$, $9 \div 1 = 9$ etc.

Also $2 \div 2 = 1$, $3 \div 3 = 1$, $5 \div 5 = 1$, $9 \div 9 = 1$ etc.

Thus, we get the following two properties:

Property 1: When a non-zero number is divided by 1, the quotient is the number itself.

Thus, $6 \div 1 = 6$, $7 \div 1 = 7$, $8 \div 1 = 8$ etc.

Property 2: When a non-zero number is divided by itself, the quotient is 1.

Thus, $6 \div 6 = 1$, $7 \div 7 = 1$, $8 \div 8 = 1$ etc.

Property 3: Dividing any number by 0 has no meaning.

Let us try to divide 3 by 0 by the use of repeated subtraction.

Given number

3

Take away 0

- 0

What remains?

3

Take away 0

- 0

What remains?

3

Take away 0

- 0

What remains?

3

⋮

and so on.

Thus, every time, we subtract 0 and get 3.

This process has no end. So, we cannot divide any number by 0.

Property 4: When 0 is divided by any non-zero number, the quotient is 0.

We know that: (any number $\times$ 0) = 0

$\therefore 2 \times 0 = 0$ gives $0 \div 2 = 0$, $3 \times 0 = 0$ gives $0 \div 3 = 0$.

$4 \times 0 = 0$ gives $0 \div 4 = 0$ and so on.

$\therefore$ When 0 is divided by any non-zero number, the quotient is 0.



Exercise 27

Fill in the blanks.

1. $9 \div 1 = \dots\dots\dots$

4. $65 \div 1 = \dots\dots\dots$

7. $7 \div \dots\dots\dots = 7$

10. $18 \div 18 = \dots\dots\dots$

13. $9 \div \dots\dots\dots = 1$

16. $0 \div 5 = \dots\dots\dots$

2. $16 \div 1 = \dots\dots\dots$

5. $\dots\dots\dots \div 1 = 8$

8. $19 \div \dots\dots\dots = 19$

11. $53 \div 53 = \dots\dots\dots$

14. $\dots\dots\dots \div 7 = 1$

17. $0 \div 9 = \dots\dots\dots$

3. $39 \div 1 = \dots\dots\dots$

6. $\dots\dots\dots \div 1 = 25$

9. $8 \div 8 = \dots\dots\dots$

12. $1 \div 1 = \dots\dots\dots$

15. $14 \div \dots\dots\dots = 1$

18. $\dots\dots\dots \div 8 = 0$



Division using Multiplication Tables (Long Division Method)

Example 1: Divide 54 by 9, using multiplication tables.

Solution:

Step 1: Arrange the numerals as shown herewith.

Step 2: Recite the multiplication table of 9 till you come to 54.

9 goes into 54 six times.

$\therefore$ Quotient = 6.

Write the quotient above the dividend, as shown.

Step 3: Now, $9 \times 6 = 54$.

Write 54 below the dividend and subtract.

$54 - 54 = 0$.

Thus, we get 0 as remainder.

$\therefore 54 \div 9 = 6$.

Here, Dividend = 54, Divisor = 9, Quotient = 6 and Remainder = 0.

Note: In such problems where remainder is 0, we always have

Divisor $\times$ Quotient = Dividend

$$\begin{array}{r} 6 \\ 9 \overline{) 54} \\ \underline{-54} \\ 0 \end{array}$$

Simple Problems on Equal Sharing

Example 2: A man distributes 27 apples among 9 children. How many apples does each child get?

Solution: Total number of apples = 27

Total number of children = 9

Here, we have to divide 27 apples into 9 groups.



So, number of apples with each child

= number of apples in each group

= $27 \div 9$

= 3.

Hence, each child gets 3 apples.

$$\begin{array}{r} 3 \\ 9 \overline{) 27} \\ \underline{-27} \\ 0 \end{array}$$



Exercise 28

Find the quotient when:

- | | | | |
|-----------------|-----------------|-----------------|-----------------|
| 1. $21 \div 3$ | 2. $24 \div 4$ | 3. $42 \div 6$ | 4. $30 \div 5$ |
| 5. $48 \div 8$ | 6. $36 \div 9$ | 7. $35 \div 7$ | 8. $63 \div 7$ |
| 9. $45 \div 5$ | 10. $64 \div 8$ | 11. $81 \div 9$ | 12. $40 \div 5$ |
| 13. $72 \div 8$ | 14. $56 \div 7$ | 15. $48 \div 6$ | 16. $36 \div 6$ |

17. A man gives 12 bananas to 4 monkeys equally. How many bananas does each monkey get?
18. Aniket has 54 metres of rope. He cuts it into 9 equal pieces. What is the length of each piece?
19. I have 25 books. I can keep 5 books in one shelf. How many shelves do I need in my almirah?
20. Pinki has 28 toys. She puts them equally in 7 boxes. How many toys are there in each box?
21. A tailor has 56 buttons. He puts 8 buttons on one shirt. How many shirts can he make with these buttons?
22. A vegetable seller has 24 kg of tomatoes. 6 men came and bought equal quantities of tomatoes. How much tomatoes did each man buy?
23. A squirrel jumps 3 steps at a time. How many jumps will it take to cover 27 steps?
24. Anu takes 4 minutes to make one chapati. How many chapatis will she make in 32 minutes?



Division without Remainder using Long Division Method

Example 1: Divide 96 by 3.

Solution:

Step 1: Arrange the numbers as shown here.

Step 2: Divide 9 tens by 3.

$$9 \text{ tens} \div 3 = 3 \text{ tens.}$$

Write 3 in tens place in the quotient.

$$\text{Now, } 3 \times 3 \text{ tens} = 9 \text{ tens.}$$

Write 9 in tens place below the dividend.

Step 3: $9 \text{ tens} - 9 \text{ tens} = 0 \text{ tens.}$

Write 0 tens as shown and bring down 6 ones.

Step 4: Divide 6 ones by 3.

$$6 \text{ ones} \div 3 = 2 \text{ ones.}$$

T	O
3	2
3	9 6
-	9
0	6
-	6
	0



Write 2 in ones place in the quotient.
Write the product $3 \times 2 = 6$ below 6 as shown.

Step 5:

Subtract: $6 - 6 = 0$.

So, we get 0 as remainder.

Thus, we have:

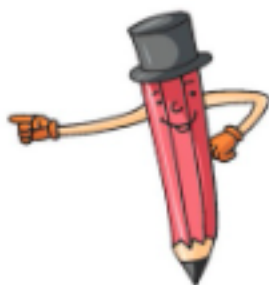
Dividend = 96, Divisor = 3, Quotient = 32 and Remainder = 0.

Check:

Divisor $\times$ Quotient = Dividend.

Here, $3 \times 32 = 96$.

Hence, the answer is verified.



Example 2: Divide 648 by 2.

Solution:

Step 1:

Arrange the numerals as shown here.

Step 2:

Divide 6 hundreds by 2.

$6 \text{ hundreds} \div 2 = 3 \text{ hundreds}$.

Write 3 in hundreds place in the quotient.

Step 3:

Bring down 4 tens. Divide 4 tens by 2.

$4 \text{ tens} \div 2 = 2 \text{ tens}$.

Write 2 in tens place in the quotient.

Step 4:

Bring down 8 ones. Divide 8 ones by 2.

$8 \text{ ones} \div 2 = 4 \text{ ones}$.

Write 4 in ones place in the quotient.

Hence, $648 \div 2 = 324$.

	H	T	O
	3	2	4
2	6	4	8
	-6		
	0	4	
		-4	
		0	8
			-8
			0

Example 3: Divide 804 by 4.

Solution:

Step 1:

Arrange the numerals as shown here.

Step 2:

Divide 8 hundreds by 4.

$8 \text{ hundreds} \div 4 = 2 \text{ hundreds}$.

Write 2 in hundreds place in the quotient.

Step 3:

Bring down 0 tens. Divide 0 tens by 4.

$0 \text{ tens} \div 4 = 0 \text{ tens}$.

Write 0 in tens place in the quotient.

Step 4:

Bring down 4 ones. Divide 4 ones by 4.

$4 \text{ ones} \div 4 = 1 \text{ one}$.

Write 1 in ones place in the quotient.

$\therefore$ Quotient = 201.

Hence, $804 \div 4 = 201$.

	H	T	O
	2	0	1
4	8	0	4
	-8		
	0	0	
		-0	
		0	4
			-4
			0

Example 4: Divide 8462 by 2.**Solution:****Step 1:** Arrange the numerals as shown below.**Step 2:** Divide 8 thousands by 2.

$$8 \text{ thousands} \div 2 = 4 \text{ thousands.}$$

Write 4 in thousands place in the quotient.

Step 3: Bring down 4 hundreds.

Divide 4 hundreds by 2.

$$4 \text{ hundreds} \div 2 = 2 \text{ hundreds.}$$

Write 2 in hundreds place in the quotient.

Step 4: Bring down 6 tens.

Divide 6 tens by 2.

$$6 \text{ tens} \div 2 = 3 \text{ tens.}$$

Write 3 in tens place in the quotient.

Step 5: Bring down 2 ones.

Divide 2 ones by 2.

$$2 \text{ ones} \div 2 = 1 \text{ one.}$$

Write 1 in ones place in the quotient.

$$\therefore \text{Quotient} = 4231.$$

$$\text{Hence, } 8462 \div 2 = 4231.$$

Th	H	T	O
4	2	3	1
2	8	4	6
	8		
	0	4	
		4	
		0	6
			6
			0
			2
			2
			0

**Exercise 29****Divide and find the quotient.**

- | | | | | |
|------------------|------------------|------------------|-------------------|-------------------|
| 1. $36 \div 3$ | 2. $64 \div 2$ | 3. $48 \div 4$ | 4. $93 \div 3$ | 5. $55 \div 5$ |
| 6. $66 \div 6$ | 7. $80 \div 4$ | 8. $50 \div 5$ | 9. $82 \div 2$ | 10. $69 \div 3$ |
| 11. $363 \div 3$ | 12. $408 \div 4$ | 13. $826 \div 2$ | 14. $936 \div 3$ | 15. $666 \div 6$ |
| 16. $604 \div 2$ | 17. $800 \div 4$ | 18. $306 \div 3$ | 19. $2468 \div 2$ | 20. $3693 \div 3$ |

Division with Regrouping

Example 1: Divide 96 by 4 and check your answer.

Solution:

Step 1: Arrange the numerals as shown here.

Step 2: Divide 9 tens by 4.

$$4 \times 2 = 8.$$

Write 2 in tens place in the quotient and 8 below 9.

$$9 \text{ tens} - 8 \text{ tens} = 1 \text{ ten}.$$

Step 3: Bring down 6 ones.

$$1 \text{ ten} + 6 \text{ ones} = 16 \text{ ones}$$

Divide 16 ones by 4.

$$4 \times 4 = 16.$$

Write 4 in ones place in the quotient and 16 below 16.

$$\text{Now, } 16 \text{ ones} - 16 \text{ ones} = 0 \text{ ones}.$$

So, we get 0 as remainder.

Thus, we have:

$$\text{Dividend} = 96, \text{ Divisor} = 4, \text{ Quotient} = 24 \text{ and Remainder} = 0.$$

Check: Divisor $\times$ Quotient = Dividend

$$\text{Here, } 4 \times 24 = 96.$$

Hence, the answer is correct.

$$\text{So, } 96 \div 4 = 24.$$

	T	O
	2	4
4)	9	6
	-8	
	1	6
	-1	6
		0



Example 2 : Divide 735 by 3.

Solution:

Step 1: Arrange the numerals as shown.

Step 2: Divide 7 hundreds by 3.

3 goes into 7 two times.

$$\text{And, } 3 \times 2 = 6.$$

Write 2 in hundreds place in the quotient and 6 below 7.

$$7 \text{ hundreds} - 6 \text{ hundreds} = 1 \text{ hundred}.$$

Step 3: Bring down 3 tens.

$$\text{Now, } 1 \text{ hundred} + 3 \text{ tens} = 13 \text{ tens}.$$

Divide 13 tens by 3.

3 goes into 13 four times.

$$\text{And, } 3 \times 4 = 12.$$

Write 4 in tens place in the quotient and 12 below 13.

$$13 \text{ tens} - 12 \text{ tens} = 1 \text{ ten}.$$

	H	T	O
	2	4	5
3)	7	3	5
	-6		
	1	3	
	-1	2	
		1	5
		-1	5
			0

Step 4:

Bring down 5 ones.

Now, 1 ten + 5 ones = 15 ones.

Divide 15 ones by 3.

3 goes into 15 five times.

And, $3 \times 5 = 15$.

Write 5 in ones place in the quotient and 15 below 15.

Now, 15 ones - 15 ones = 0 ones.

$\therefore$ Remainder = 0.

Thus, Quotient = 245 and Remainder = 0.

Likewise, 4-digit numbers may also be divided.

Example 3: Divide 9051 by 7.**Solution:**

	Th	H	T	O
	1	2	9	3
7)	9	0	5	1
	-7			
	2	0		
	-1	4		
		6	5	
		-6	3	
			2	1
			-2	1
				0

$$\therefore 9051 \div 7 = 1293.$$

**Example 4: Divide 384 by 6.****Solution:****Step 1:**

Arrange the numerals as shown.

Step 2:

First, we start with hundreds.

6 does not go into 3.

$\therefore$ We start with tens.

3 hundreds + 8 tens = 38 tens.

And, $6 \times 6 = 36$.

Write 6 in tens place in the quotient and 36 below 38.

38 tens - 36 tens = 2 tens.

	H	T	O
	6	4	
6)	3	8	4
	-3	6	
		2	4
		-2	4
			0

Step 3:

Bring down 4 ones.

$2 \text{ tens} + 4 \text{ ones} = 24 \text{ ones.}$

Divide 24 ones by 6.

6 goes into 24 four times.

And, $6 \times 4 = 24.$

Write 4 in the quotient and 24 below 24.

$24 \text{ ones} - 24 \text{ ones} = 0 \text{ ones.}$

$\therefore \text{Remainder} = 0.$

Thus, Quotient = 64.

Hence, $384 \div 6 = 64.$

**Example 5: Divide 7504 by 8.****Solution:**

Th	H	T	O
	9	3	8
8	7	5	0 4
	-7	2	
	3	0	
	-2	4	
	6	4	
	-6	4	
		0	

$$\therefore 7504 \div 8 = 938.$$

**Example 6: Divide 9045 by 9.****Solution:**

Th	H	T	O
	1	0	0 5
9	9	0	4 5
	-9		
	0	0	
	-0		
	0	4	
	-0		
	4	5	
	-4	5	
		0	

$$\therefore 9045 \div 9 = 1005.$$





Exercise 30

Divide and find the quotient.

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| 1. $57 \div 3$ | 2. $60 \div 4$ | 3. $75 \div 5$ | 4. $68 \div 4$ |
| 5. $90 \div 6$ | 6. $84 \div 7$ | 7. $95 \div 5$ | 8. $96 \div 8$ |
| 9. $81 \div 3$ | 10. $76 \div 4$ | 11. $84 \div 6$ | 12. $92 \div 4$ |
| 13. $314 \div 2$ | 14. $972 \div 4$ | 15. $630 \div 5$ | 16. $744 \div 6$ |
| 17. $920 \div 8$ | 18. $805 \div 7$ | 19. $756 \div 7$ | 20. $414 \div 3$ |
| 21. $5601 \div 3$ | 22. $8064 \div 7$ | 23. $6175 \div 5$ | 24. $9432 \div 8$ |
| 25. $6138 \div 6$ | | | |



Divide and find the quotient.

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| 26. $356 \div 4$ | 27. $225 \div 9$ | 28. $510 \div 6$ | 29. $776 \div 8$ |
| 30. $711 \div 9$ | 31. $3276 \div 7$ | 32. $4504 \div 8$ | 33. $9051 \div 7$ |
| 34. $7136 \div 8$ | 35. $5706 \div 9$ | | |

Word Problems

Example 1: A bottle of coke costs ₹ 6. How many bottles of coke can be bought for ₹ 90?

Solution: Total cost of bottles purchased = ₹ 90
 Cost of one bottle = ₹ 6
 Number of bottles purchased = $90 \div 6$
 = 15

Hence, 15 bottles can be bought for ₹ 90.

T	O
1	5
6	90
-6	↓
3	0
-3	0
	0



Example 2: A boy is cycling at the rate of 8 km per hour. How long does he take to cover 192 km?

Solution:

Total distance to be covered = 192 km.

Distance covered in 1 hour = 8 km.

$$\begin{aligned}\therefore \text{Time taken} &= (192 \div 8) \text{ hours} \\ &= 24 \text{ hours.}\end{aligned}$$

Hence, the boy takes 24 hours to cover 192 km.

	H	T	O
	2	4	
8)	1	9	2
	-1	6	
		3	2
		-3	2
			0



Example 3: The product of two numbers is 686. If one of them is 7, find the other.

Solution:

Product of two numbers = 686.

One number = 7.

$$\begin{aligned}\text{The other number} &= 686 \div 7 \\ &= 98.\end{aligned}$$

Hence, the other number is 98.

	H	T	O
	9	8	
7)	6	8	6
	-6	3	
		5	6
		-5	6
			0

Example 4: 5202 nails are packed equally in 9 boxes. How many nails are there in each box?

Solution:

Number of nails in 9 boxes = 5202.

$$\begin{aligned}\text{Number of nails in 1 box} &= 5202 \div 9 \\ &= 578.\end{aligned}$$

Hence, number of nails in each box = 578.

	Th	H	T	O
	5	7	8	
9)	5	2	0	2
	-4	5		
		7	0	
		-6	3	
			7	2
			-7	2
				0

Example 5: A poultry farm produces 3115 eggs in one week.
How many eggs are produced each day?

Solution:

We know that 1 week = 7 days.

Number of eggs produced in 7 days = 3115.

Number of eggs produced each day

$$= 3115 \div 7$$

$$= 445.$$

Hence, the number of eggs produced each day is 445.



	Th	H	T	O
		4	4	5
7)	3	1	1	5
	-2	8		
	3	1		
		-2	8	
		3	5	
			-3	5
				0



Exercise 31

1. From a ribbon of length 56 cm, how many pieces each 4 cm long can be cut?
2. 6 toy guns cost ₹ 108. What is the cost of one toy gun?
3. There are 7 days in a week. How many weeks are there in 112 days?
4. 162 books are arranged on 9 shelves. Each shelf contains the same number of books. How many books are there on each shelf?
5. 504 students were divided into 6 equal groups. How many students are there in each group?



6. A pen costs ₹ 8. How many such pens can be had for ₹ 768?
7. 9 T-shirts cost ₹ 864. What is the cost of one T-shirt?
8. 6 students can sit on a bench. How many benches are needed for 852 students?
9. 7 carts can carry 525 boxes of apples. How many boxes can each cart carry?
10. How many times can you take away 8 from 368?



11. A man distributed ₹ 1101 equally among his three sons. How much money did each son get?

12. A car goes 8 km in one litre of petrol. How much petrol will it consume to go 1040 km?

13. The cost of 7 bicycles is ₹ 6041. Find the cost of one bicycle.

14. The price of a notebook is ₹ 9. How many notebooks can be purchased for ₹ 7713?

15. The product of two numbers is 7304. If one of them is 8, what is the other number?



Division with Remainder

Note here that in the process of division, once we get a remainder which is less than the divisor, we cannot continue the process any further.

Example 1: Divide 237 by 6 and check the result.

Solution: We have:

	H	T	O
	3	9	
6	2	3	7
-	1	8	
	5	7	
-	5	4	
			3



Here, $3 < 6$, i.e., remainder is less than divisor, we cannot continue the process further.

∴ Quotient = 39 and Remainder = 3.

Check: Dividend = 237, Divisor = 6, Quotient = 39 and Remainder = 3.

Always remember that:

$$\text{Divisor} \times \text{Quotient} + \text{Remainder} = \text{Dividend.}$$

$$\text{Now, Divisor} \times \text{Quotient} + \text{Remainder} = 6 \times 39 + 3$$

$$= 234 + 3$$

$$= 237$$

$$= \text{Dividend.}$$

Hence, the answer is correct.



Example 2: Divide 3709 by 8 and check the result.**Solution:** We have:

	Th	H	T	O
	4	6	3	
8)	3	7	0	9
	-3	2		
		5	0	
		-4	8	
			2	9
			-2	4
				5

 $\therefore$ Quotient = 463 and Remainder = 5.**Check :** Here, Dividend = 3709, Divisor = 8, Quotient = 463 and Remainder = 5.

$$\begin{aligned}
 \text{Now, Divisor} \times \text{Quotient} + \text{Remainder} &= 8 \times 463 + 5 \\
 &= 3704 + 5 \\
 &= 3709 \\
 &= \text{Dividend.}
 \end{aligned}$$

Hence, the answer is correct.

Division by 10**Example 3: Find the quotient and remainder when:**

(a) $68 \div 10$

(b) $253 \div 10$

(c) $1637 \div 10$

What do you conclude?

Solution: We have:

(a)

	T	O
	6	
10)	6	8
	-6	0
		8

 $\therefore$ Quotient = 6,
Remainder = 8

(b)

	H	T	O
	2	5	
10)	2	5	3
	-2	0	
		5	3
		-5	0
			3

 $\therefore$ Quotient = 25,
Remainder = 3

(c)

	Th	H	T	O
	1	6	3	
10)	1	6	3	7
	-1	0		
		6	3	
		-6	0	
			3	7
			-3	0
				7

 $\therefore$ Quotient = 163,
Remainder = 7**Conclusion:** When a number is divided by 10, the quotient is obtained by removing the ones digit from the number. The digit at ones place is the remainder.



Exercise 32

Find the quotient and remainder. Also check your answer in each case.

- | | | | |
|-------------------|-------------------|--------------------|--------------------|
| 1. $537 \div 2$ | 2. $407 \div 3$ | 3. $159 \div 4$ | 4. $278 \div 5$ |
| 5. $476 \div 3$ | 6. $591 \div 4$ | 7. $796 \div 9$ | 8. $700 \div 8$ |
| 9. $487 \div 6$ | 10. $479 \div 9$ | 11. $453 \div 7$ | 12. $701 \div 8$ |
| 13. $615 \div 9$ | 14. $1023 \div 7$ | 15. $2109 \div 6$ | 16. $3514 \div 7$ |
| 17. $4756 \div 8$ | 18. $5938 \div 9$ | 19. $2307 \div 4$ | 20. $5213 \div 6$ |
| 21. $347 \div 10$ | 22. $514 \div 10$ | 23. $1037 \div 10$ | 24. $2416 \div 10$ |

More Word Problems

Example: 700 students are divided equally into 6 groups. How many students are there in each group? How many are leftover?

Solution:

Total number of students = 700

Number of groups = 6

Number of students in each group = $700 \div 6$

On dividing 700 by 6, we get 116 as quotient and 4 as remainder.

Hence, there are 116 students in each group and 4 students are leftover.

	H	T	O
	1	1	6
6)	7	0	0
	-6		
	1	0	
		-6	
		4	0
			-3
			6
			4



Exercise 33

- How many groups of 10 children each can be made from a class of 43 children? How many children will be leftover?
- How many bundles of 12 pencils each can be formed from 100 pencils? How many pencils are leftover?
- 96 trees are to be planted. If 7 trees are planted in each row, how many rows are formed? How many plants are leftover?



4. At the most 6 girls can sit on a bench. What is the minimum number of benches required for 45 girls to sit?

[Hint : 45 when divided by 6 gives quotient = 7, remainder = 3; 7 benches are occupied and 3 girls remain, who can sit on 1 bench.]

5. 108 exercise books were divided equally among 10 boys. How many will each get and how many exercise books remain?

6. There are 7 days in a week and 365 days in a year. How many weeks are there in a year? How many days are left?

7. A pen costs ₹ 6. A man had ₹ 1154 with him with which he bought as many pens as he could. How many pens did he buy and how much money is left with him?

8. The bookshop gives 9 notebooks to each student. If there are 1272 notebooks in the shop, how many students can get the notebooks? How many notebooks are left?

9. 1878 saplings were bought to be planted in rows of 8 saplings each. How many rows are formed and how many saplings are left?

10. 4764 bulbs are to be packed equally in 7 cartons. How many bulbs are there in each carton? How many bulbs are left?



Things to Remember

1. In a division sum $a \div b = c$, we call 'a' as dividend, 'b' as divisor and 'c' as quotient.
2. When a non-zero number is divided by 1, the quotient is the number itself.
3. When a non-zero number is divided by itself, the quotient is 1.
4. Division by zero is meaningless.
5. When 0 is divided by any non-zero number, the quotient is 0.
6. Divisor $\times$ Quotient + Remainder = Dividend.
7. When a number is divided by 10, the quotient is obtained by removing the ones digit from the number. The digit at ones place is the remainder.

**QUESTION BAG 1****(Objective Type Questions)**

Tick (✓) the correct answer.

1. Division is a process of repeated
(a) addition ☐ (b) subtraction ☐
(c) multiplication ☐ (d) None of these ☐
2. Divide 81 by me, the answer is me. Divide 54 by me it's 6 you will see. Who am I?
(a) 6 ☐ (b) 7 ☐ (c) 8 ☐ (d) 9 ☐
3. Dividend =
(a) remainder $\times$ quotient + divisor ☐ (b) quotient $\times$ divisor + remainder ☐
(c) quotient + divisor $\times$ remainder ☐ (d) none of these ☐
4. How many times should 6 be subtracted from 48 to give 0?
(a) 4 ☐ (b) 6 ☐ (c) 7 ☐ (d) 8 ☐
5. 6 eggs are needed to bake a cake. If Mrs Roy uses 66 eggs, how many cakes has she baked?
(a) 9 ☐ (b) 10 ☐ (c) 11 ☐ (d) 12 ☐
6. $135 \div 135 = ?$
(a) 0 ☐ (b) 1 ☐ (c) 2 ☐ (d) 135 ☐
7. $0 * 12 = 0$
This will be true if * is replaced by
(a) $\times$ or $\div$ ☐ (b) $+$ or $\div$ ☐ (c) $+$ or $-$ ☐ (d) $\times$ or $+$ ☐
8. Sanchit has 47 oranges. If he puts them equally in 5 baskets, how many oranges will be leftover?
(a) 2 ☐ (b) 3 ☐ (c) 5 ☐ (d) 7 ☐
9. If 94 is divided by 4, the remainder is
(a) 0 ☐ (b) 2 ☐ (c) 4 ☐ (d) None of these ☐
10. The quotient of $256 \div 8$ is
(a) 24 ☐ (b) 27 ☐ (c) 28 ☐ (d) 32 ☐
11. The dividend in the expression $240 \div 6 = 40$ is
(a) 0 ☐ (b) 6 ☐ (c) 40 ☐ (d) 240 ☐
12. Out of the following divisions, one has a different quotient. Which is it?
(a) $12 \div 4$ ☐ (b) $24 \div 8$ ☐ (c) $36 \div 9$ ☐ (d) $15 \div 5$ ☐

13. Out of the following divisions, one has a different quotient. Which is it?
 (a) $63 \div 9$ ☐ (b) $48 \div 6$ ☐ (c) $32 \div 4$ ☐ (d) $64 \div 8$ ☐
14. Difference of 1000 and 200, when divided by 4 gives
 (a) 400 ☐ (b) 200 ☐ (c) 300 ☐ (d) 250 ☐
15. There are 8 chairs at each table in a restaurant. There are 96 chairs in all. Which number sentence can be used to find the number of tables in the restaurant?
 (a) $8 \times 96 = \dots$ ☐ (b) $96 - \dots = 8$ ☐ (c) $8 + \dots = 96$ ☐ (d) $96 \div 8 = \dots$ ☐
16. In which of the following cases do we obtain 1 as quotient?
 (a) Number divided by itself ☐ (b) Number divided by 0 ☐
 (c) 1 divided by a number ☐ (d) Number divided by 1 ☐
17. How many 8s are there in 120?
 (a) 12 ☐ (b) 15 ☐ (c) 40 ☐ (d) 41 ☐
18. The number of weeks in 91 days is
 (a) 7 ☐ (b) 13 ☐ (c) 17 ☐ (d) 73 ☐
19. Sonal has ₹ 50. A pen costs ₹ 7 each. What is the greatest number of pens Sonal can buy?
 (a) 6 ☐ (b) 7 ☐ (c) 8 ☐ (d) 9 ☐
20. A man has ₹ 58 with him. What is the maximum number of 5 rupee coins he can have?
 (a) 10 ☐ (b) 11 ☐ (c) 12 ☐ (d) None of these ☐
21. Last week, 228 grade 3 students rode on 4 buses on a field trip. The same number of students rode on each bus. How many students rode on each bus?
 (a) 52 ☐ (b) 57 ☐ (c) 62 ☐ (d) 67 ☐
22. Rajan has 4 plates of apples as shown below.
- 



- If Rajan puts the apples equally in the four plates, what should be the number of apples in each plate?
 (a) 3 ☐ (b) 4 ☐ (c) 5 ☐ (d) 6 ☐
23. When the largest 3-digit number is divided by 3, we get
 (a) 33 ☐ (b) 303 ☐ (c) 333 ☐ (d) 111 ☐
24. If 25 people contribute ₹ 3 each for the purchase of 5 articles each of the same price, what is the price of each article?
 (a) ₹ 5 ☐ (b) ₹ 15 ☐ (c) ₹ 20 ☐ (d) ₹ 25 ☐
25. The remainder obtained, when 420 is divided by 8 is
 (a) 0 ☐ (b) 1 ☐ (c) 2 ☐ (d) 4 ☐

QUESTION BAG 2

1. Fill in the blanks:

- (a) The answer of division is called the
- (b) The number which divides is called the
- (c) If a number is by itself, the answer is 1.
- (d) If $3 \times 350 = 1050$, then $1050 \div 3 = \dots\dots\dots$

2. State whether each of the following statements is true or false:

- (a) Division is equal distribution.
- (b) If $32 \div 4 = 8$, then $4 \times 8 = 32$
- (c) If $40 \div 10 = 4$, then $4 \div 10 = 40$
- (d) Any number divided by 0 is the number itself.
- (e) In $27 \div 3 = 9$, 9 is the dividend.
- (f) If 4 is the quotient and 8 is the divisor, then the dividend is 12.
- (g) We can change the order of numbers in division.
- (h) Dividend is always greater than or equal to the divisor.
- (i) The remainder is always smaller than the divisor.
- (j) We start the division process from the ones place.
- (k) $0 \div$ any non-zero number $= 0$
- (l) Any number $\div 0 = 0$

3. Fill in the divisor:

- | | | |
|-----------------------------------|-----------------------------------|-----------------------------------|
| (a) $56 \div \dots\dots\dots = 8$ | (b) $45 \div \dots\dots\dots = 9$ | (c) $72 \div \dots\dots\dots = 9$ |
| (d) $36 \div \dots\dots\dots = 6$ | (e) $24 \div \dots\dots\dots = 6$ | (f) $64 \div \dots\dots\dots = 8$ |
| (g) $42 \div \dots\dots\dots = 7$ | (h) $49 \div \dots\dots\dots = 7$ | (i) $63 \div \dots\dots\dots = 7$ |
| (j) $30 \div \dots\dots\dots = 6$ | | |

4. Fill in the dividend:

- | | |
|----------------------------------|------------------------------------|
| (a) $\dots\dots\dots \div 9 = 5$ | (b) $\dots\dots\dots \div 9 = 9$ |
| (c) $\dots\dots\dots \div 9 = 2$ | (d) $\dots\dots\dots \div 8 = 4$ |
| (e) $\dots\dots\dots \div 6 = 8$ | (f) $\dots\dots\dots \div 5 = 8$ |
| (g) $\dots\dots\dots \div 7 = 8$ | (h) $\dots\dots\dots \div 10 = 10$ |

5. How many flowers are there in each vase, if we put:

- (a) 18 flowers in 3 vases (b) 28 flowers in 7 vases
(c) 45 flowers in 5 vases (d) 30 flowers in 6 vases
(e) 81 flowers in 9 vases (f) 56 flowers in 8 vases

6. Fill in the blanks:

- (a) 63 days make weeks.
(b) 48 pencils make dozens.
(c) A necklace has 11 beads. So necklaces have 110 beads.
(d) If 27 coins are equally placed in 3 bags, there are coins in each bag.
(e) If 8 persons travel in a jeep then jeeps are needed for 56 persons.
(f) Arushi distributes 54 chocolates among her 9 friends. Then each friend gets chocolates .

7. The missing addends in each problem are all the same. Find the missing numbers:

- (a) + + + + = 40
(b) + + + = 16
(c) + + + = 28
(d) + + + + + + = 63
(e) + + + + + + = 35

8. Fill in the blanks with <, > or = :

- (a) $16 \div 8$ $27 \div 3$ (b) $14 \div 2$ $45 \div 5$
(c) $36 \div 4$ $18 \div 3$ (d) $28 \div 4$ $35 \div 5$
(e) $24 \div 3$ $28 \div 4$ (f) $12 \div 3$ $24 \div 6$

9. Fill in the blank boxes:

- (a) $15 \div 15 = \square$ (b) $10 \div 1 = \square$ (c) $6 \div 6 = \square$ (d) $0 \div 7 = \square$
(e) $18 \div 1 = \square$ (f) $4 \div 4 = \square$ (g) $9 \div \square = 1$ (h) $8 \div \square = 8$
(i) $\square \div 2 = 0$ (j) $\square \div 1 = 79$

10. For a division sum, answer the following:

- (a) If 7 is the divisor, can we have 6 as remainder?
- (b) If 9 is the divisor, can we have 9 as remainder?
- (c) If 2 is the divisor, can we have 4 as remainder?

11. Complete the following table:

		Dividend	Divisor	Quotient	Remainder
(a)	$76 \div 8$				
(b)	$38 \div 6$				
(c)	$54 \div 5$				
(d)	$30 \div 4$				
(e)	$83 \div 9$				
(f)	$67 \div 7$				
(g)	$73 \div 9$				

12. Complete the following table:

		Quotient	Remainder
(a)	$46 \div 10$		
(b)	$379 \div 10$		
(c)	$700 \div 10$		
(d)	$7776 \div 10$		
(e)	 $\div 10$	2	3
(f)	 $\div 10$	34	7
(g)	 $\div 10$	56	0
(h)	 $\div 10$	83	5

13. Fill in the blanks:

- (a) There are eights in 33 and will be leftover.
(b) There are fives in 36 and will be leftover.
(c) There are sixes in 40 and will be leftover.
(d) There are fours in 34 and will be leftover.



- 14.** A fruit seller had to arrange 992 apples in 8 boxes. How many apples were there in one box?
15. How many times is 1736 than 4?

16. Fill in the missing digit(s) in the quotient.

(a)
$$\begin{array}{r} \square 2 \\ 8 \overline{) 256} \end{array}$$

(b)
$$\begin{array}{r} \square 5 \\ 7 \overline{) 665} \end{array}$$

(c)
$$\begin{array}{r} 8 \square \\ 6 \overline{) 522} \end{array}$$

(d)
$$\begin{array}{r} \square 8 \square \\ 2 \overline{) 568} \end{array}$$

(e)
$$\begin{array}{r} 1 \square 3 \\ 5 \overline{) 815} \end{array}$$

(f)
$$\begin{array}{r} \square 6 \square \\ 4 \overline{) 664} \end{array}$$

(g)
$$\begin{array}{r} 6 \square \\ 9 \overline{) 621} \end{array}$$

(h)
$$\begin{array}{r} 1 \square 7 \\ 5 \overline{) 935} \end{array}$$

- 17.** Divide the smallest 4-digit number by 8.
18. Divide the greatest 3-digit number by 7. Find the quotient and remainder.
19. Find the quotient.

(a) $504 \div 3$

(b) $364 \div 7$

(c) $670 \div 5$

(d) $774 \div 6$

(e) $7488 \div 8$

(f) $8953 \div 7$

(g) $7578 \div 9$

(h) $8487 \div 9$

20. Find the quotient and remainder.

(a) $594 \div 4$

(b) $447 \div 6$

(c) $889 \div 7$

(d) $755 \div 9$

(e) $2908 \div 6$

(f) $5051 \div 7$

(g) $8076 \div 9$